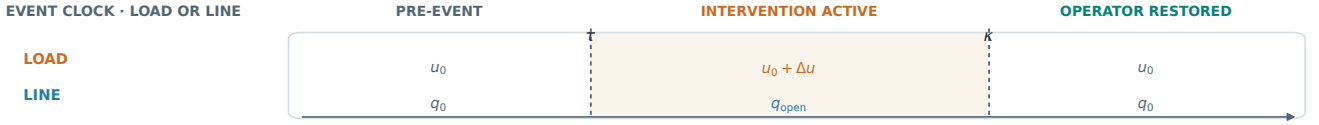
FROM SWITCHED PHYSICS TO CAUSAL PMU REPORT

A restored input changes the network mode; the physical and measurement states keep evolving.



01 SWITCHED PLANT

$$\dot{x} = f_q(x, z, u)$$

$0 = g_q(x, z, u)$

load input or branch mode changes
differential and algebraic states

02 ELECTRICAL MAP

$$h_k = H_q(x, z, u)$$

V phasors at PMU buses
oriented I at declared ends
terminal assignment is fixed

03 CAUSAL FILTER

$$\eta_k = F_M(\eta_{k-1}, h_k)$$

phase history is carried forward
phase evolution yields *f*
filter history yields ROCOF

04 REPORT TO INFERENCE

$$y_k = G_M(\eta_k, h_k) + b + \varepsilon_k$$

8 PMUs · 30 reports/s
47 causal coordinates

V · 16 | I · 15 | f/R · 16

AT RESTORATION, THE OPERATOR RESETS; MEMORY DOES NOT

DYNAMIC STATE x
$x(\kappa^+) = x(\kappa^-)$
machine and control response continues

NETWORK STATE z
$g_{q_0}(x, z, u_0) = 0$ is solved again
voltages and terminal currents are recomputed

FILTER STATE η
η carries phase history forward
post-restoration reports retain the transient

INFERENCE INPUT $y_{1:k}$ only · no true source, switch time, solver state, or raw waveform

47 coordinates: 8 |V| + 8 angle(V) + 8 |I| + 7 angle(I) + 8 f + 8 ROCOF.

Phase-A phasors are exported; B/C phases are reconstructed. Raw waveforms and harmonics are not observed.